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The Hong Kong University of Science
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[Senali Withanachchi](#)

Education

PhD Environmental Engineering

(Sep 2026 – Present)

The Hong Kong University of Science
and Technology, Hong Kong
Major: Environmental Engineering

BSc (Hons) Plant Biotechnology

(Jan 2020 – June 2024)

University of Colombo, Sri Lanka
Major: Biotechnology
Class: First Class Honours

BSc (Hons) Psychology

(Jul 2021 – Jul 2022)

University of Northampton, UK

Skills

- Soil-Based and Soilless Plant Cultivation
- Biochemical & Metabolic Assays
- Molecular Biology & Gene Expression
- Analytical & Technical Diagnostics
- Microbiological Techniques

Curriculum Vitae

Senali Withanachchi

PhD researcher in Environmental Engineering, specializing in the application of nanobubble technology to optimize plant growth in hydroponic, aquaponic, and bioponic systems. Holds a BSc (Hons) First Class degree in Plant Biotechnology from the University of Colombo, Sri Lanka. Research focuses on understanding the underlying mechanisms of nanobubbles by investigating their effects on biochemical pathways, plant metabolism, and gene expression, contributing to the advancement of modern agriculture and the development of sustainable environmental solutions.

Professional Experience

Sep 2026 – Present

Graduate Research Assistant

Dept. of Civil & Environmental Engineering

The Hong Kong University of Science and Technology,
Hong Kong

- Research Focus: Nanobubble Technology in Hydroponics/ Aquaponics/ Bioponics: Mechanistic Insights on Biochemical Activity and Gene Expression
- PI: Prof. Samir Kumar Khanal

Jun 2025 – Jul 2026

Lecturer

Royal Institute Campus, Sri Lanka-

University of London / King's College London Programs

- Course Area: Advanced Research Design and Statistics

Aug 2024 – Jun 2025

Graduate Teaching Assistant

Department of Plant Sciences, University of Colombo, Sri Lanka

Jun 2022 – Jun 2024

Undergraduate Research Student

Department of Plant Sciences, University of Colombo, Sri Lanka

- Research Focus: Medicinal Plants and High-Value Crops Biochemistry; Nitrogen Metabolism; Waste Valorization and Resource Recovery; Hydroponics and Organic Fertilizers
- PI: Dr. Pradeepika Saputhanthri

Awards and Honors

1. **HKUST RedBird PhD Recruitment Award**–The Hong Kong University of Science and Technology – (Sep 2026–Current)
2. **PostGraduate Studentship**–The Hong Kong University of Science and Technology – (Sep 2026–Current)
3. **Recipient of the Dean's Lists** – University of Colombo, SL – (2019/2020-2023/2024)
4. **The Best Presenter Award** – 44th Annual Sessions of the Institute of Biology, SL (IOBSL) (Plant Sciences and Agriculture) – (Sep 2024)
5. **The Best Presenter Award** – 29th International Forestry and Environmental Symposium, University of Sri Jayawardenepura, SL (Forest and natural resources + Biodiversity conservation and management session) – (Jan 2025)
6. Lalith Athulathmudali Mahapola Higher Education Scholarship from the trust fund managed by the University Grants Commission, SL (2020-2024)
7. Delegate –HPAIR Harvard (University of Harvard) and Asia (Hong Kong) Conferences (2023)
8. Next Generation Women Leaders Asia Pacific –McKinsey (2022)
9. Championship, Scrabble League–University of Colombo SL (2023-2024)

Conference Presentations (* denotes presenter)

1. **Withanachchi, S. V***, Saputhanthri, P. S., In-vitro antioxidant, sun protection, anti-inflammatory, and antibacterial activity of *F. leucopyrus*, *O. octandra*, and *H. speciosa*. 29th International Forestry & Environmental Symposium, 18 Jan 2025, Sri Lanka, Oral Presentation.
2. **Withanachchi, S. V***, Saputhanthri, P. S., Antioxidant and sun protection capacity of Ceylonblack tea brew based on tea grades and manufacturing technique. Proc. 29th International Forestry & Environmental Symposium, 18 Jan 2025, Oral Presentation.
3. **Withanachchi, S. V***, Saputhanthri, P. S. (2024). Antioxidant and sun protection activity of selected medicinal plants in Sri Lanka. Proc. IOBSL 44th Annual Sessions, 27 Sep 2024, Sri Lanka. Oral Presentation
4. **Withanachchi, S. V***, Saputhanthri, P. S. (2024). Waste flowers as a sustainable source for cosmetic production. Proc. IOBSL 44th Annual Sessions, 27 Sep 2024, Sri Lanka. Oral Presentation.
5. **Withanachchi, S. V.**, et al. (2024). Development of an organic fertilizer to replace commercial hydroponic Albert solution to cultivate “Kankung” plants. Proc. IOBSL 44th Annual Sessions, 27 Sep 2024, Poster Presentation.

Certifications

1. **Conducting Life Cycle Assessments using LCA Software Tools** (According to ISO 14040:2006 & 14044:2006 standard, SimaPro software with ecoinvent database) - National Cleaner Production Centre, SL (Established by UNIDO under the Ministry of Industry and Commerce, SL) - (Apr 2026)
2. **Consultant Development in Carbon Footprint Quantification Based on ISO 14064-1:2018, ISO 14064-2:2019 & ISO 14067:2018 Standards** - National Cleaner Production Centre, SL - (Nov 2025)
3. **Career Development Program in Nanotechnology and Advanced Characterization**- Sri Lanka Institute of Nanotechnology- SLINTEC -(Aug-Sep 2025)
4. **Summer Training in Research Methodologies**- Delhi University, India - (Jun-Jul 2021)